

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I SK. Arshad Basha (192312336) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SK. Arshad Basha

20+20+22+19+13
= 94
MP

Q10

disaster-response agency deploys AI-guided drones to airlift medical supplies to villages cut off by a wildfire.

An AI-guided drone must deliver medical supplies to villages isolated by wildfire. The environment is uncertain because roads and trails can become blocked suddenly, smoke changes visibility and movement cost.

i) Greedy Best-First Search (GBFS)

Greedy Best-First search selects the next location using only the heuristic

$$f(n) = h(n)$$

$\therefore h(n)$ is the estimated straight-line distance from the current location to the village

* It ignores the actual cost already spent reaching that location.

ii) A* would handle the same situation

A* combines the actual cost with the heuristic:

$$f(n) = g(n) + h(n)$$

$\therefore g(n)$ = actual cost from starting point to node n

$h(n)$ = estimated remaining cost from n to the village

$f(n)$ = estimated total route cost.

Advantages

- * considers both distance and actual travel cost.
- * can incorporate terrain, smoke, battery consumption.
- * with an admissible heuristic, provide an optimal least-cost path

Example

Route	Distance estimate $h(n)$	Current cost $g(n)$	$f(n) = g(n) + h(n)$
Route A	5 km	12	17
Route B	7 km	5	12

iii) Impact of Heuristic Accuracy

The heuristic is very important for both algorithms

Greedy Best First search

- * depends almost completely on $h(n)$
- * An inaccurate heuristic can cause the drone to choose a poor route
- * If straight-line distance does not account for smoke or blocked trails, GBFS can be highly misleading.

Thus:

GBFS: highly depended on $h(n)$

A* : balance $g(n)$ & $h(n)$

A well-designed heuristic can significantly reduce a search time.

of dynamic changes

every blocked trails create a dynamic search environment.

If a trail becomes blocked after the drone has planned its route

GBFS:-

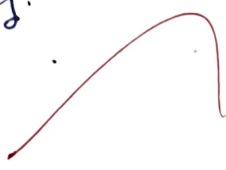
- * The old path may become invalid.
- * The drone must update the search and choose another route.
- * Because GBFS focuses only on the heuristic, it may again choose a route that looks close but has high risk.

A* generally increase:-

- * The affected path costs can be updated or removed from search graph
- * A* can recalculate an alternative path.

conclusion

For an emergency drone, A* provides the best balance between speed, route cost and safety.

$$f(n) = g(n) + h(n)$$


2) Scenario

A large agriculture enterprise an AI system to optimize water distribution through a network of smart irrigation valves across thousands of acres.

The agriculture irrigation system can be formulated as a large-scale optimization problem.

i) Hill climbing as a local search algorithm that starts with initial value configuration and repeatedly moves toward as better neighboring configuration.

working

1. start with an initial value configuration
2. measure its objective value
3. change one or more value settings slightly
4. Evaluate the new configuration.
5. If the new configuration is better, move to it.

$x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow x_3$

ii) A configuration may be better than all of its immediate neighbors but worse than another configuration elsewhere

* The system adjusts several nearby values and finds a configuration that saves water.

* However, reaching a much better configuration requires the temporality increase water usage in one zero.

Increasing value

$v_1 \& v_2$

iii) simulated Annealing (SA)

simulated Annealing improves Hill climbing by sometimes accepting a worse solution

The probability of accepting a worse solution is approximately.

$$= e^{-\Delta E / T}$$

ΔE = increase in objective cost

T = Temperature

Here $T \rightarrow 0$

iv) recommended Approach

For thousands of irrigation values, I would recommend a Genetic Algorithm combined with real-time sensor-based optimization possibly followed by local refinement.

Why Genetic Algorithm?

- * Thousands of decision variables
- * complex interactions between values
- * dynamic weather conditions
- * multiple competing objective.

sensor data $\rightarrow$ GA optimization $\rightarrow$ Local Refinement.

3) Scenario

An Autonomous submarine is deployed to explore an uncharted ocean trench. It must navigate safely to map the trench floor, avoid hazards.

i) Why an online search Agent is required.

- * An online search agent is suitable because the submarine does not have a complete map of uncharted ocean trench before starting its mission.

Example

Observe $\rightarrow$ plan $\rightarrow$ move $\rightarrow$ observe $\rightarrow$ update

- * Thermal vents
- * deep trenches
- * narrow passages
- * unexpected obstacles.

ii) Challenges of partial observability

* The submarine's sonar has a limited range, so it cannot observe the entire environment.

current location + nearby obstacles.

- * currents
- * Temperature variations
- * sediment movements
- * changing visibility.

iii) Building and updating the Internal model

* The submarine maintains an internal representation of the environment.

- * starting position
- * mission destination/area
- * Basic navigation information
- * known hazards, if any.

Sonar $\rightarrow$ Rock detected at (x, y)

$m_{new} = \text{update}(m_{old}, \text{sensor data})$

* safe / free

search vs uninformed & informed search

Algorithms such as BFS, DFS and UCS search without domain-specific heuristic information.

$$f(n) = g(n) + h(n)$$

where

$g(n)$ = is the cost already travelled

$h(n)$ = estimate the remaining cost.

* online search allows submarines to interleave planning
search $\leftrightarrow$ Action $\leftrightarrow$ observation

v) most suitable Approach.

* The best choice is an online informed search approach with the continuous sensing & replanning

A practical strategy would combine

online search + sonar mapping + Heuristics

$$f(n) = g(n) + h(n) + R(n)$$

where :-

$g(n)$ = travel cost

$h(n)$ = estimated remaining distance

$R(n)$ = risk associated with the route.

sense $\rightarrow$ update map $\rightarrow$ search $\rightarrow$ Act.